BHUKAVACH: AI-ASSISTED UNDERGROUND MINE SAFETY, MONITORING AND RESCUE ROVER

METHODOLOGY

BHUKAVACH is a proposed rugged, remotely controlled tracked rover that can help mine rescue and safety teams understand underground conditions before entering a hazardous area. The rover will be equipped with an HD camera and lights for clear live viewing, a thermal sensor to detect heat sources, gas sensors to monitor O_2, CH_4 and CO_2, and environmental sensors to measure temperature, humidity and pressure. LiDAR will be used to measure distances and understand the surrounding area. An ESP32-based onboard system will collect the sensor data and send it to a surface control station. The rover will be operated remotely by a human, while AI will assist in analysing camera and thermal data, identifying possible hazards and observing changes in gas levels. The final decisions will remain with the rescue team. Since communication inside underground mines can be difficult, the prototype will also be tested for communication reliability, sensor performance, battery life and operation in challenging environments.

PROPOSED BLOCK DIAGRAM

UNDERGROUND ENVIRONMENT
(Hazards, Survivor, Terrain)
↓
BHUKAVACH ROVER
(HD Camera, Thermal, Gas, Environmental, LiDAR)
↓
ONBOARD ELECTRONICS
(ESP32, Data Acquisition, Power & Status)
↓
WIRELESS COMMUNICATION LINK
(Live Video, Sensor Data, Automated reporting)
↓
SURFACE CONTROL STATION
(Remote Operation, Live Monitoring, Alerts)
↓
AI-ASSISTED ANALYSIS
(Visual/Thermal, Gas Trends, Hazard Detection)
↓
MISSION REPORT
(Recorded Evidence, Sensor Data, AI-Assisted Summary)

PROPOSED OUTCOME

BHUKAVACH provides a practical platform for remotely inspecting underground mine areas and monitoring conditions in real time. It provides live camera, thermal and environmental data to the operator, helps identify possible hazards, records important mission data and supports the generation of a post-mission report. The system focuses on remote operation, allowing rescue teams to gather information about underground conditions before human entry. Its key functions include monitoring gas levels, temperature and humidity, observing heat sources, understanding the surrounding area and identifying possible hazards through AI-assisted analysis. The system is designed to support safer and more informed rescue operations, while keeping all important operational decisions under human control.